250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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A novel urban vegetation mapping approach for fire risk assessment: A Mediterranean case study

Maria Mahamed | L. Wittenberg | H. Kutiel | A. Brook

Environmental Science, Geography · Urban Ecosystems · 24 May 2023

A series of urban **wildfires** that engulfed the Mediterranean in recent decades raised the problem of introducing urban vegetation into **fire management**. While the **Wildland**-Urban Interface (WUI) concept… Expand

3 | Springer Nature | Save | Cite

Pervasive shifts in forest dynamics in a changing world

N. McDowell | C. Allen | +20 authors | Chonggang Xu

Environmental Science · Science · 28 May 2020

TLDR The authors show that shifts in **forest dynamics** are already occurring, **and** the emerging pattern is that global **forests** are tending toward younger **stands** with faster turnover as old-growth **forest** with stable **dynamics** are dwindling. Expand

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Uncertainty Aware Wildfire Management

Tina Diao | Samriddhi Singla | Ayan Mukhopadhyay | Ahmed Eldawy | Ross D. Shachter | Mykel J. Kochenderfer

Environmental Science, Engineering · AI4SG@AAAI Fall Symposium · 15 October 2020

TLDR This paper proposes a decision-theoretic approach to combat **wildfires** by **model** the resource allocation problem as a partially-observable Markov decision process, **and** presents a data-driven **model** that lets us simulate how **fires spread** as a function of relevant covariates. Expand

9 | [PDF] | arXiv | Save | Cite

Predicting wildfire spread and behaviour in Mediterranean landscapes

M. Salis B. Arca +13 authors D. Spanò Environmental Science · 27 October 2016

The use of spatially explicit **fire spread models** to assess **fire propagation** and behaviour has several applications for **fire management** and research. We used the FARSITE simulator to predict the… Expand

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Towards predictive data-driven simulations of wildfire spread - Part II: Ensemble Kalman Filter for the state estimation of a front-tracking simulator of wildfire spread

M. Rochoux C. Emery S. Ricci B. Cuenot A. Trouvé

Environmental Science · 28 May 2014

TLDR Results indicate that data-driven **simulations** are capable of correcting inaccurate **predictions** of the **fire** front location **and** of subsequently providing an optimized **forecast** of the **wildfire behavior** at future lead times. Expand

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Simulating post-wildfire forest trajectories under alternative climate and management scenarios.

Alicia Azpeleta Tarancón P. Fulé Kristen L. Shive C. Sieg A. S. Meador Barbara A. Strom

Environmental Science · Ecological Applications · 1 October 2014

TLDR Substantial differences in **model** outputs depending on climate **and management** actions are shown, suggesting **forest** managers should incorporate climate change into the process of analyzing the environmental effects of alternative actions. Expand

46 PDF PubMed Save Cite

DATA-DRIVEN SIMULATIONS OF WILDFIRE SPREAD AT REGIONAL SCALES

Cong Zhang Environmental Science, Engineering · 2018

Title of dissertation: DATA-DRIVEN **SIMULATIONS** OF **WILDFIRE SPREAD** AT REGIONAL SCALES Cong Zhang Doctor of Philosophy, 2018 Dissertation directed by: Professor Arnaud Trouvé Department of Mechanical… Expand

2 Publisher Save Cite

Generation and evaluation of ensemble simulations of wildfire spread for probabilistic forecast

Frédéric Allaire J. Filippi V. Mallet Environmental Science ·

9 November 2018

TLDR This work proposes to perform ensemble evaluation considering all the burn probabilities that are predicted at the scale of a specific fire event and requires the observed burned surfaces corresponding to the specific event. Expand

2 PDF Publisher Save Cite

A Survey on Rainfall Prediction using Artificial Neural Network

D. Nayak Amitav Mahapatra Pranati Mishra

Environmental Science, Computer Science · 26 June 2013

TLDR A survey of available literature of some methodologies employed by different researchers to utilize ANN for rainfall prediction reports that rainfall prediction using ANN technique is more suitable than traditional statistical and numerical methods. Expand

149 Publisher Save Cite

West Coast Forecast Challenges and Development of Atmospheric River Reconnaissance

F. M. Ralph F. Cannon +16 authors L. Delle Monache

Environmental Science, Geography ·
Bulletin of The American Meteorological Society... · 1 August 2020

Water management and flood control are major challenges in the western United States. They are heavily influenced by atmospheric river (AR) storms that produce both beneficial water supply and... Expand

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